

SCHOOL OF SCIENCE

Department of Physics

PHY 1163 PHYSICS FOR ENGINEERS I: Continuous Assessment Test no 1

Level of Study: Year 1 Groups: ETE, CS, IS, CSE, ELE and MEE

Date: 23/11/2017

Duration: 2h

Maximum marks: 40

Instructions:

1. The paper contains **FOUR (4)** questions. Attempt ALL questions.
2. Write all answers in the booklet provided and use both sides of paper
3. **Start each question on a fresh page**
4. Use the last two pages of your booklet as a rough draft
5. **Do not write any answers on this question paper.**
6. Do not forget to write your registration number
7. ***It is not permitted to pass or attempt to pass any information or instrument such as pen, calculator, ruler.... from one candidate to another.***
8. You may use the following information if needed: Acceleration due to the gravity $g = 9.8\text{m/s}^2$

QUESTION 1 (10 MARKS)

- (a) Give the ***fundamental physical quantities*** which are subjected to measurement in Mechanics. **(3 marks)**
- (b) A rectangular plate has a ***length*** of $(24.5 \pm 0.2)\text{cm}$ and a width of $(10.4 \pm 0.2)\text{cm}$, those have been measured using the same ***meter stick***, as instrument. Determine the ***indirect measurements*** of the ***area*** for the plate, expressed both with the ***absolute*** and ***relative uncertainties***. Is that meter stick ***appropriate*** to measure indirectly that area? ***Justify*** your answer. **(5 marks)**
- (c) Verify whether or not the following equation is ***dimensionally consistent*** knowing that v is the speed (expressed in $\text{m} \cdot \text{s}^{-1}$), a the acceleration ($\text{m} \cdot \text{s}^{-2}$), x the position (in m), and t the time (expressed in s): $x = \frac{v^2}{2a}$ **(2 marks)**

QUESTION 2 (10 MARKS)

(a) A bus is moving with a constant deceleration. Its speed changes from $24\text{ m}\cdot\text{s}^{-1}$ to $16\text{ m}\cdot\text{s}^{-1}$ while the distance it covers is 50 m .

2. a.1. What is **the distance** it will cover before stopping? (3 marks)

2.a.2. What is **the time** needed to stop considered from the moment when its velocity is $24\text{ m}\cdot\text{s}^{-1}$. (1 mark)

(b) A ball is launched vertically upward with an initial velocity of $12\text{ m}\cdot\text{s}^{-1}$ from the roof top located at 40 m from the ground (**you are strongly asked to consider the origin of y -axis just at the ground level**). Find:

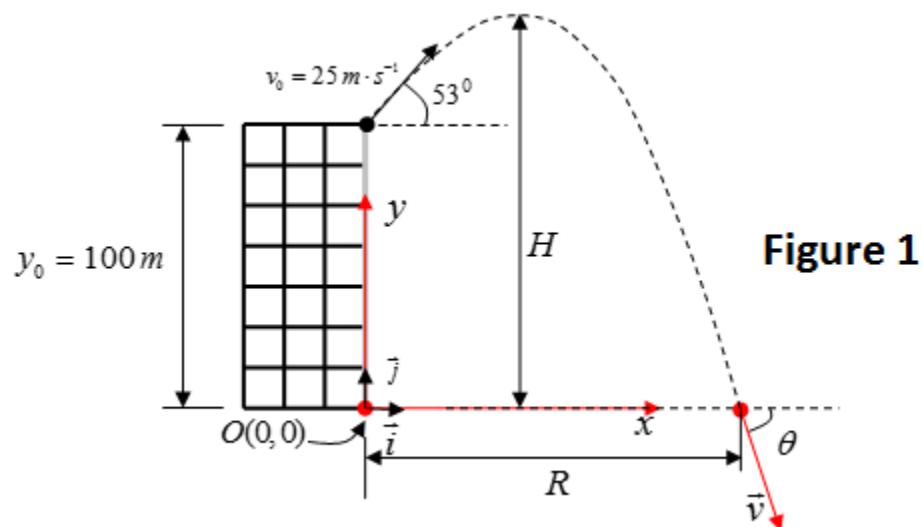
2.b.1. its velocity when just it hits the ground; (3 marks)

2.b.2. the duration of its trajectory; (1 mark)

2.b.3. its maximum height (2 marks)

QUESTION 3 (10 MARKS)

A stone is launched from the roof top of a building that is 100 m tall with the speed $v_0 = 25\text{ m}\cdot\text{s}^{-1}$, and at an angle of elevation $\theta_0 = 53^\circ$ with respect to the horizontal line as shown in Figure 1.



If we consider the origin $O(0,0)$ of the axes at the ground level, determine:

(a) the **time** needed for the stone to hit the ground; (2 marks)

- (b) its maximum *height*; (2 marks)
- (c) its *horizontal range*; (2 marks)
- (d) its *velocity* when it hits the ground; and (2 marks)
- (e) its *impact angle* at the ground. (2 marks)

QUESTION 4 (10 MARKS)

The horizontal force $\vec{F}$ of Figure 2 accelerates the block of 4 kg at $1\text{ m}\cdot\text{s}^{-2}$ to the left. It is given that the coefficient of kinetic friction is $\mu_k = 0.5$. Knowing that the block of 5 kg is ascending the slope (Figure 2), determine:

- (a) the magnitude of the force $\vec{F}$; (5 marks)
- (b) the magnitude of the tension force in the rope; (2 marks)
- (c) If $F = 10\text{ N}$ and if the block of 5 kg is descending the slope, what is the magnitude of the acceleration of the two blocks? (3 marks)

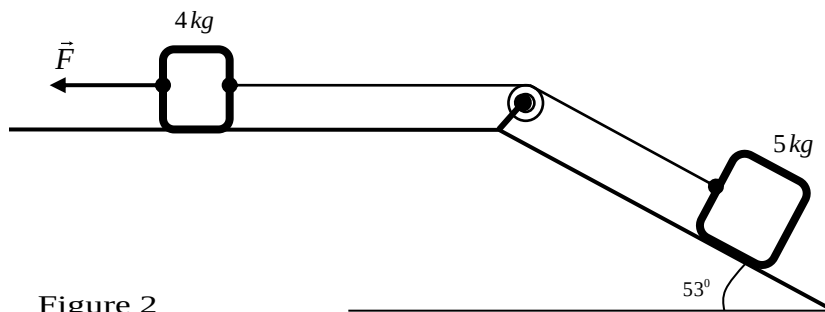


Figure 2

Good luck!